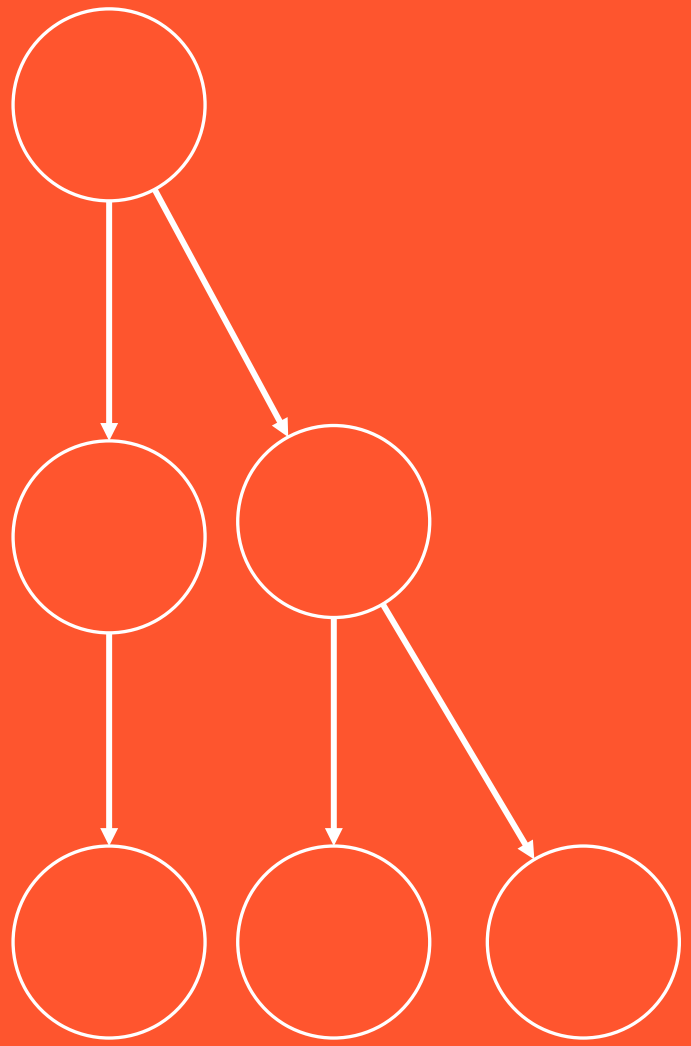


BCOG 100

Spatial Cognition

Lecture 9.3 — The Hippocampus and Its Neighbors

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: What Would You Look For?

In your group, three minutes.

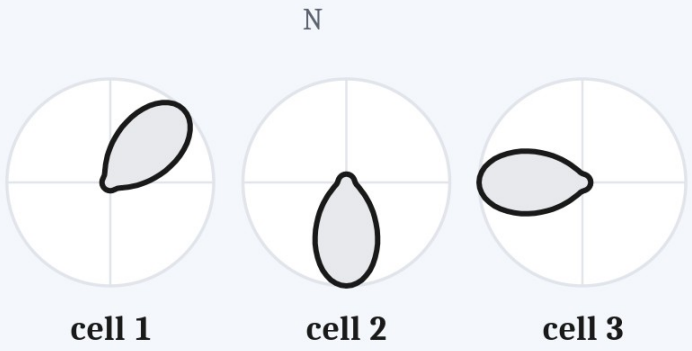
Every procedure in the second lecture could run in any substrate. Pick one: a ring attractor for heading, a running sum of position, or repeating codes for where the animal is.

If a brain ran it, what would one cell's firing look like as an animal moved around a room?

A Compass in the Brain

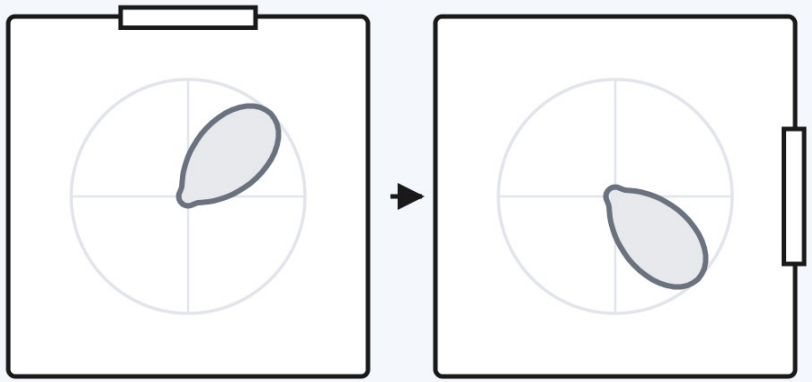
A **head-direction cell** fires when the animal's head points in one direction, whatever it is doing and wherever it is. Turns keep the signal going; stable landmarks correct it.

Three head-direction cells



Each fires for one way the head points, whatever the animal is doing and wherever it is. Together they cover the whole circle.

Rotate the card

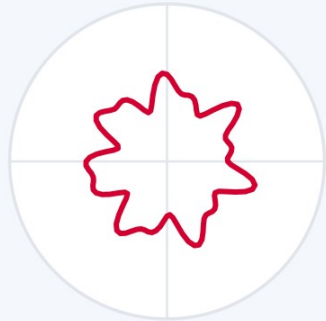


card
the north wall

card moved
the east wall

Every cell's preferred direction rotates with the card, all together.
(Drawn at 90 degrees for illustration.)

Without the turn signal



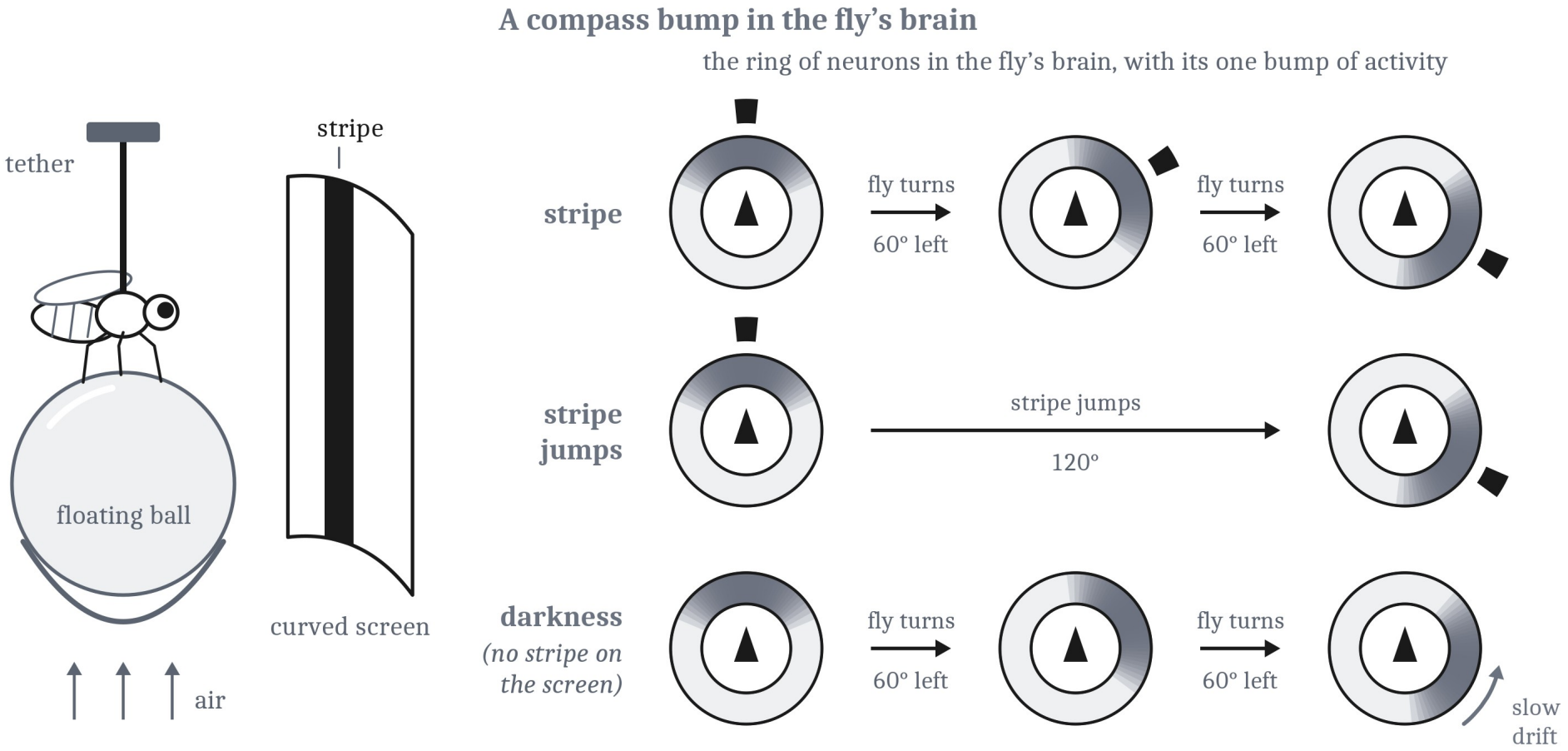
vestibular organs destroyed

Cells keep firing, but not for any one heading, and the direction signal never comes back, even with a familiar card in view.

Turns update one heading signal; stable landmarks correct it.

Rings Seen in Tissue

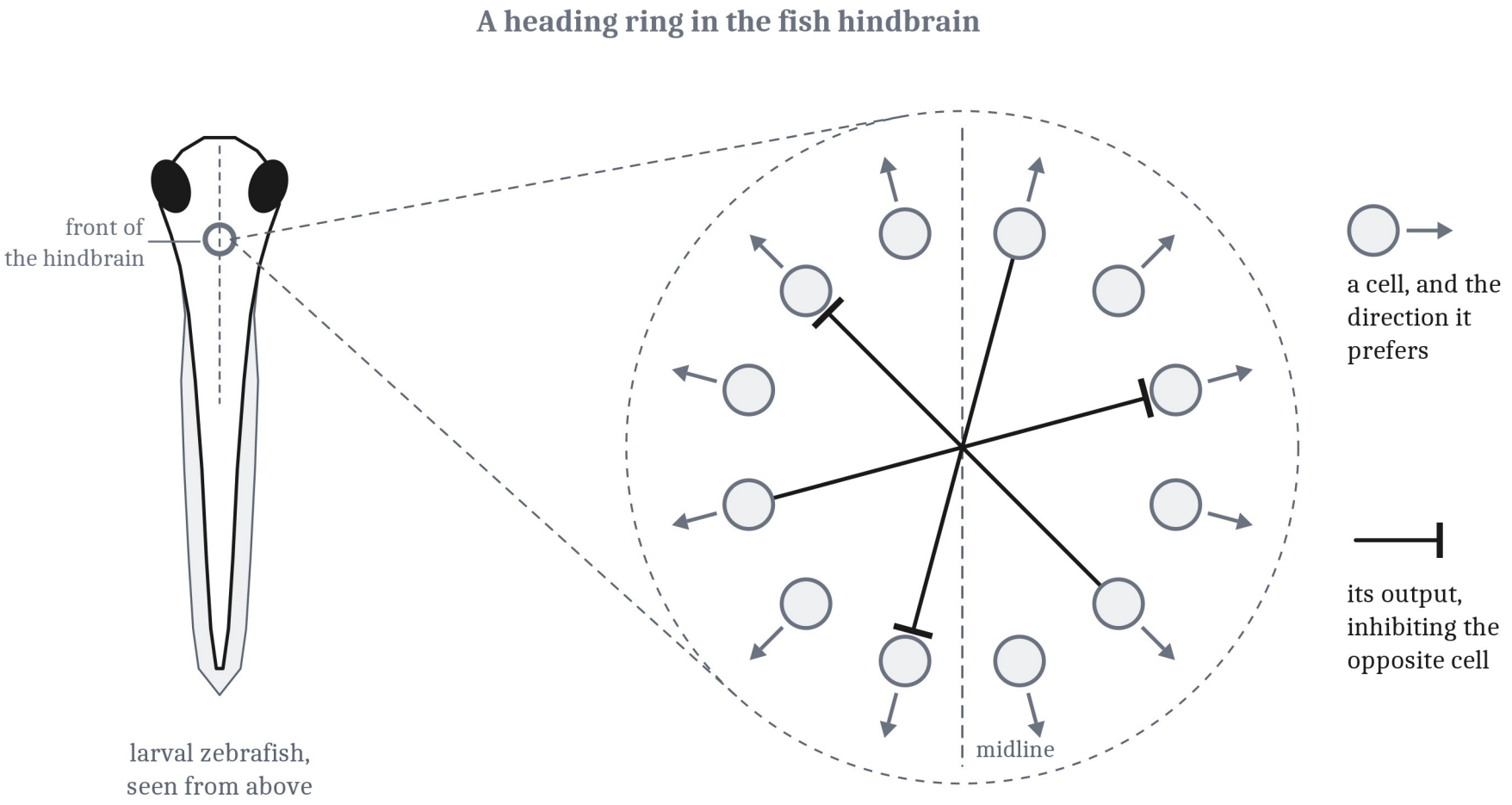
In the fruit fly and the larval zebrafish, the ring attractor described in the second lecture can be seen in the tissue. One bump of activity turns as the animal turns.



dark bar outside a ring: where the fly sees the stripe; triangle: the way the fly faces; shaded patch: the bump
schematic, after Seelig and Jayaraman (2015); bump about 82° wide

Rings Seen in Tissue

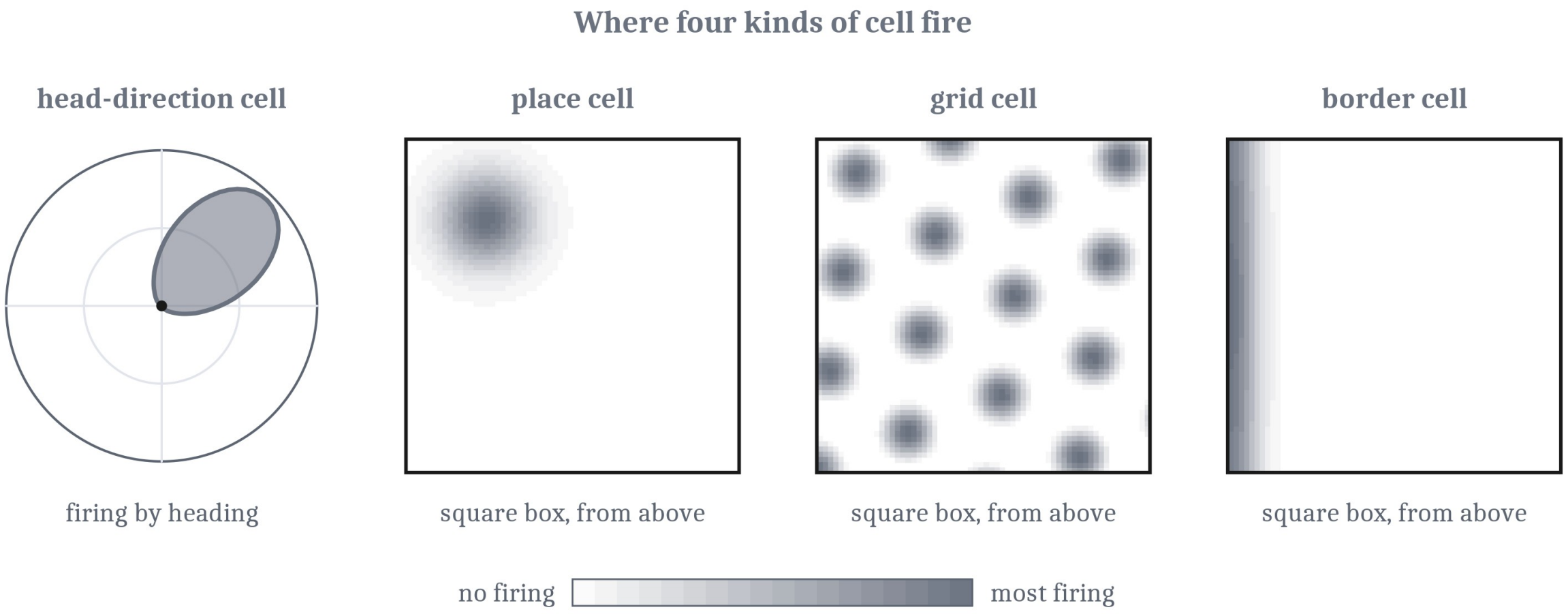
In the fruit fly and the larval zebrafish, the ring attractor described in the second lecture can be seen in the tissue. One bump of activity turns as the animal turns.



our drawing, adapted from Petrucco et al. (2023), Fig. 6e (CC BY 4.0)

Place Cells, and What Happens Without Them

A **place cell** in the hippocampus fires when the animal is in one part of its surroundings, its **place field**, whichever way it faces. Place learning depends on the hippocampus; response learning, on the striatum.

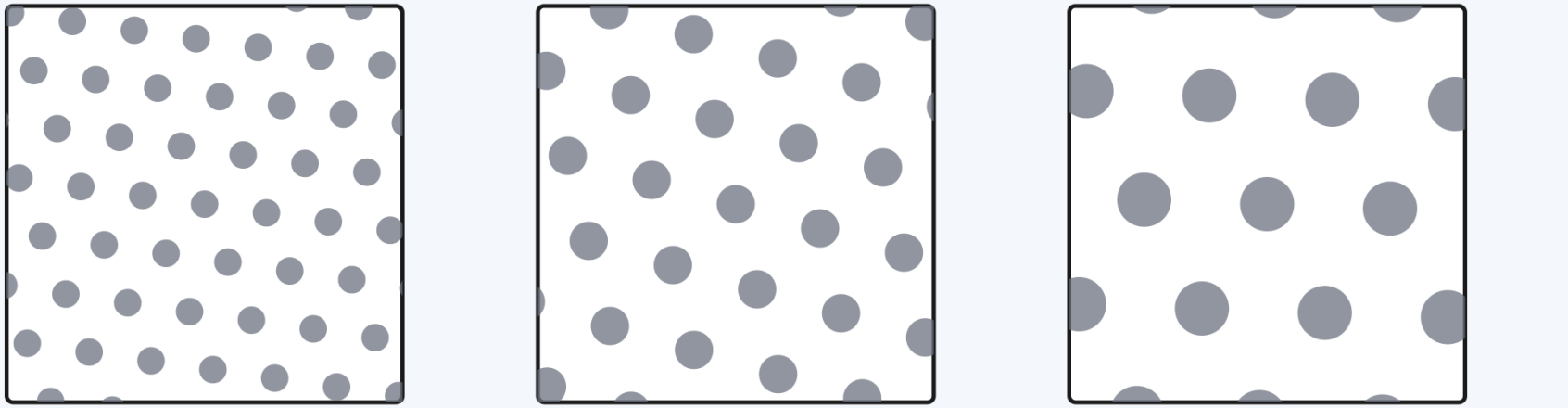


computed patterns, not recordings: each shape is the idealized form of what the cell type does

Grid Cells and Border Cells

In the **entorhinal cortex**, beside the hippocampus and feeding it, a **grid cell** fires at many places, at the corners of equilateral triangles. A **border cell** fires along a wall.

Grid cells fall into a few groups



group 1
smallest spacing

group 2
about 1.4 times wider

group 3
about 1.4 times wider again

spacing grows in steps, from the top of the entorhinal cortex to the bottom

A border cell



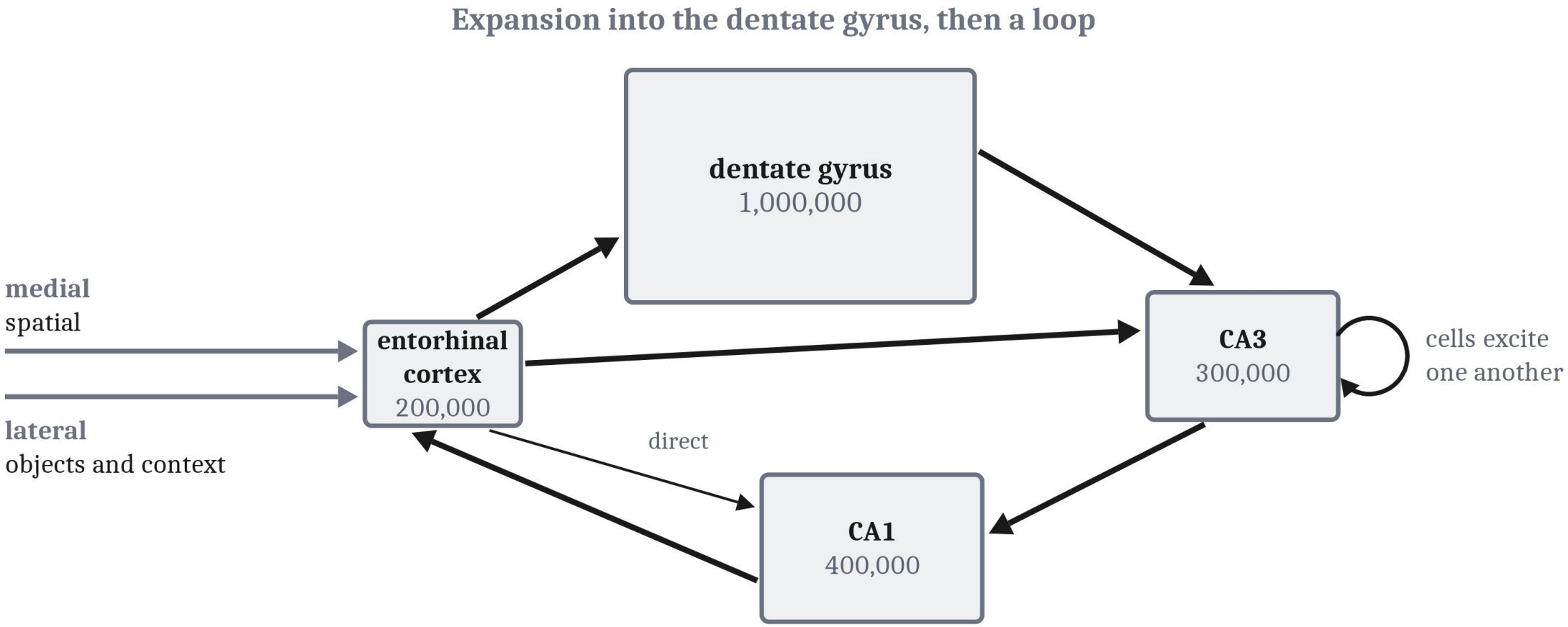
fires along one wall

Touching a wall
corrects grid drift.

Grids appear within minutes in a new place, and survive the removal of landmarks.

The Circuit

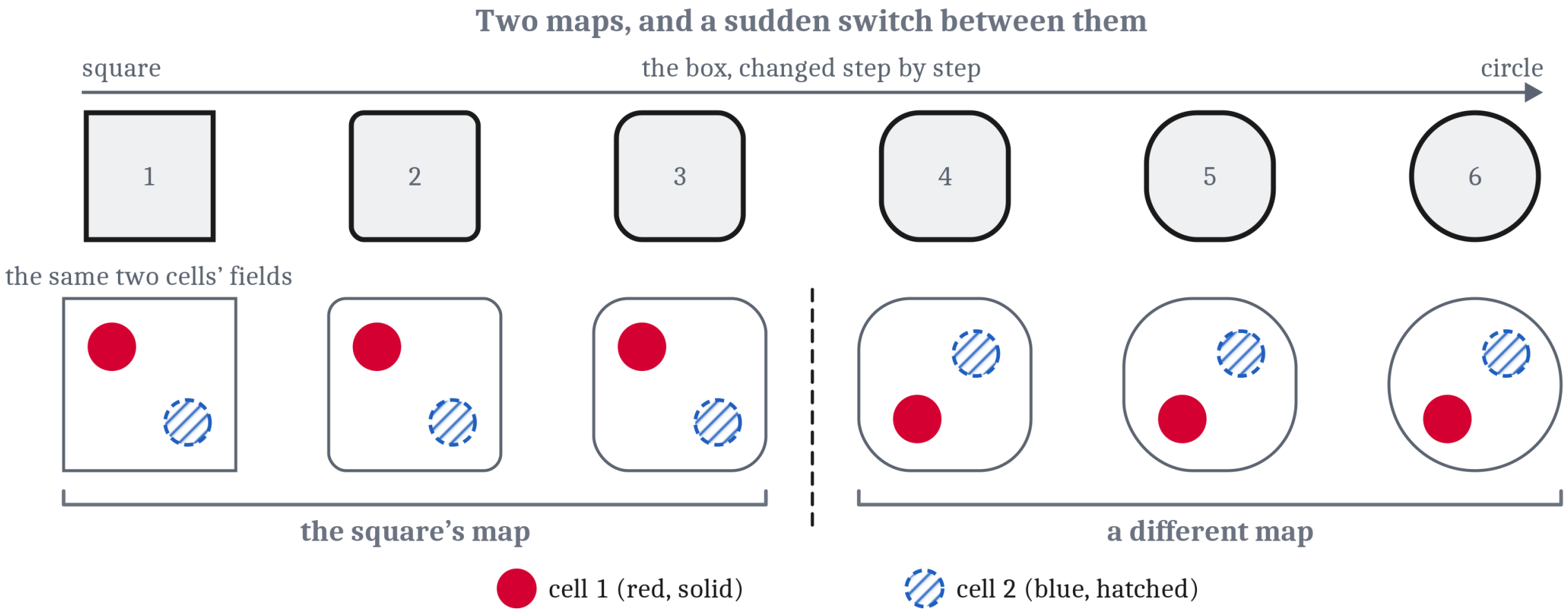
The **dentate gyrus** spreads entorhinal input across many more cells, few active at a time, and passes it to **CA3**, whose cells excite one another. **CA1** receives CA3's output and sends the result back to the entorhinal cortex.



numbers: approximate cells in each region of the rat; box areas drawn in proportion
counts from Burgess and O'Keefe (2003), after McNaughton (1989)

Many Maps from One Set of Cells

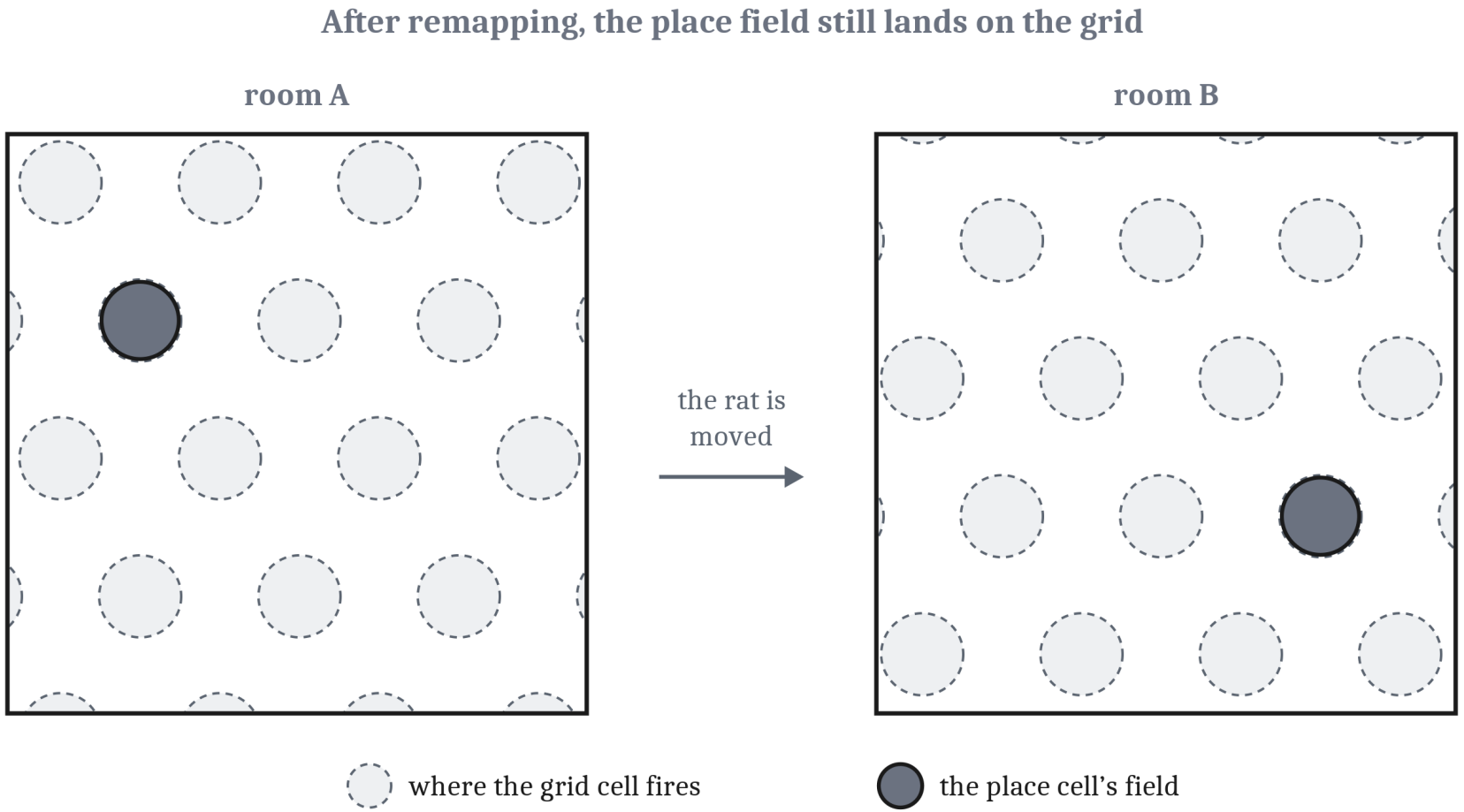
Remapping: in a new room, place cells reorganize all at once. **Pattern separation**, making overlapping inputs less alike, is thought to be the dentate gyrus's job.



schematic, after the morph-box experiment of Wills et al. (2005); field positions illustrative

Many Maps from One Set of Cells

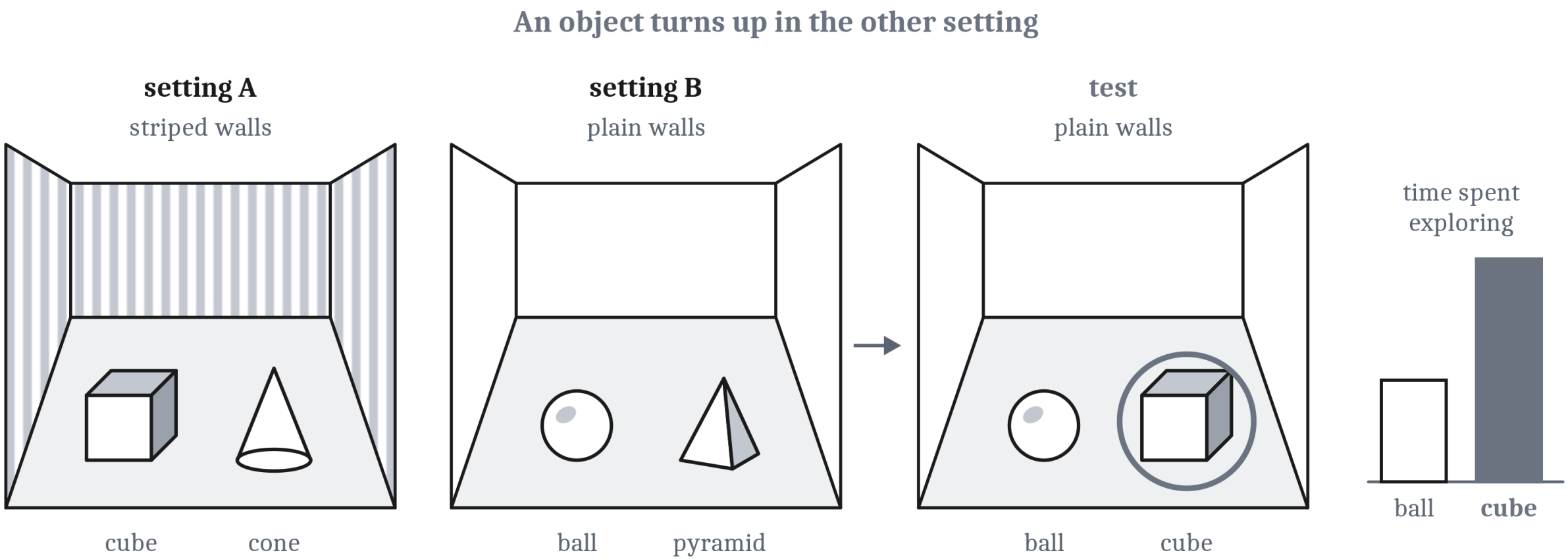
Remapping: in a new room, place cells reorganize all at once. **Pattern separation**, making overlapping inputs less alike, is thought to be the dentate gyrus's job.



our drawing, adapted from Whittington et al. (2020), Fig. 6A (CC BY 4.0)

Which Thing Was Where, and What Is New

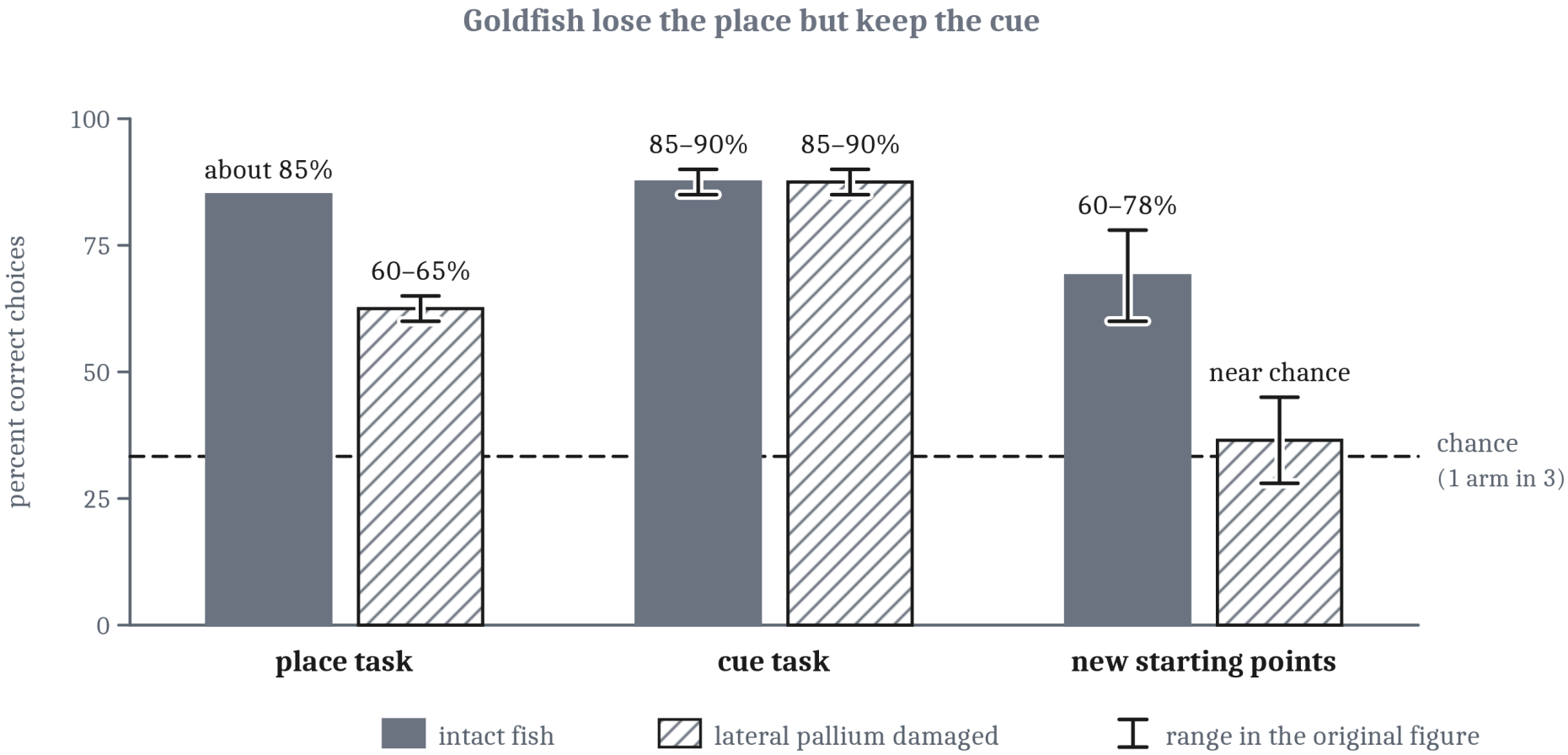
The hippocampus codes little about objects in themselves, and much about the place and context an object was in. It is also needed to learn from exploring.



Our own schematic of the object-in-context test (after Norman and Eacott, 2005); bar heights are not data

Older Than Mammals

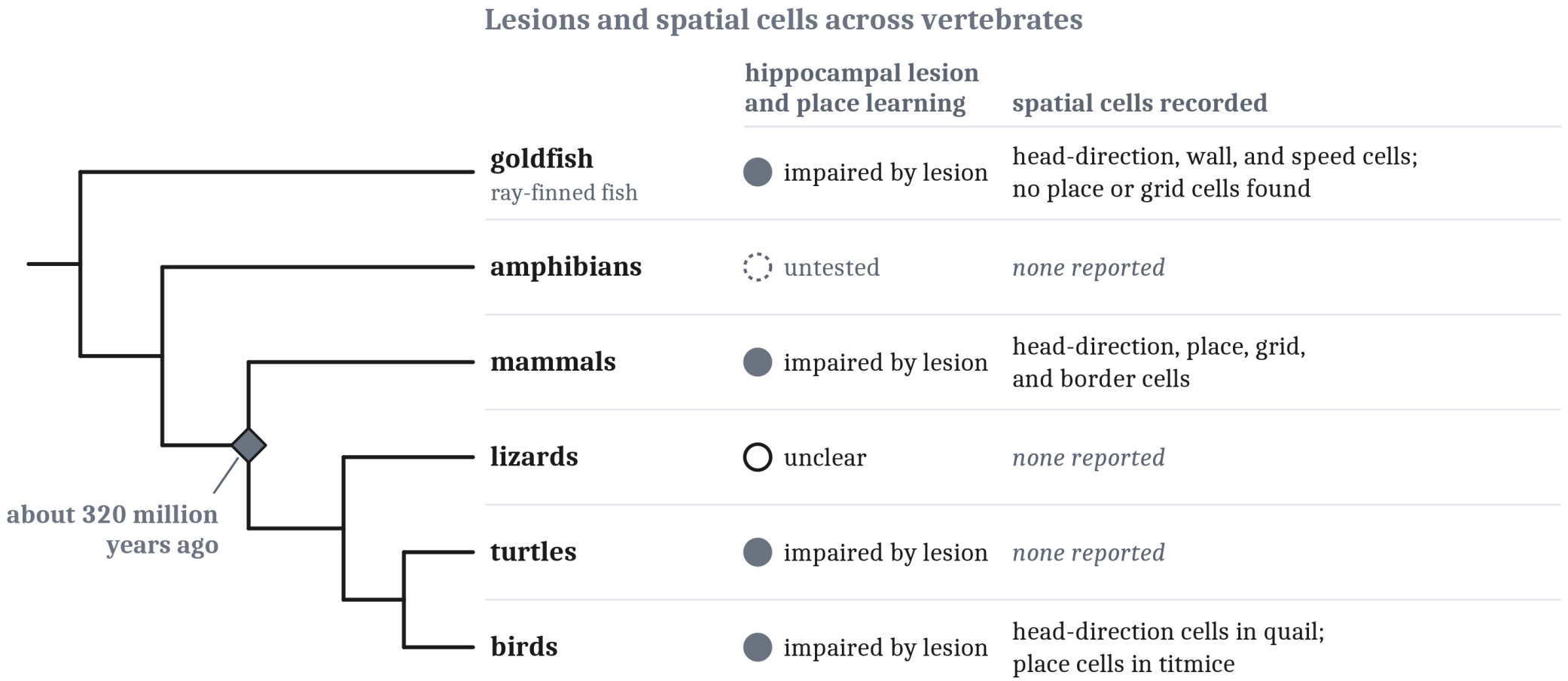
In goldfish, turtles, birds, and rats, damage to a corresponding forebrain region removes the place strategy and spares cue-following: evidence of homologs doing the same job.



Redrawn from Broglio and colleagues (2015), Fig. 4, CC BY 4.0; values read from their figure, bars at the middle of each range

What Is New in Mammals

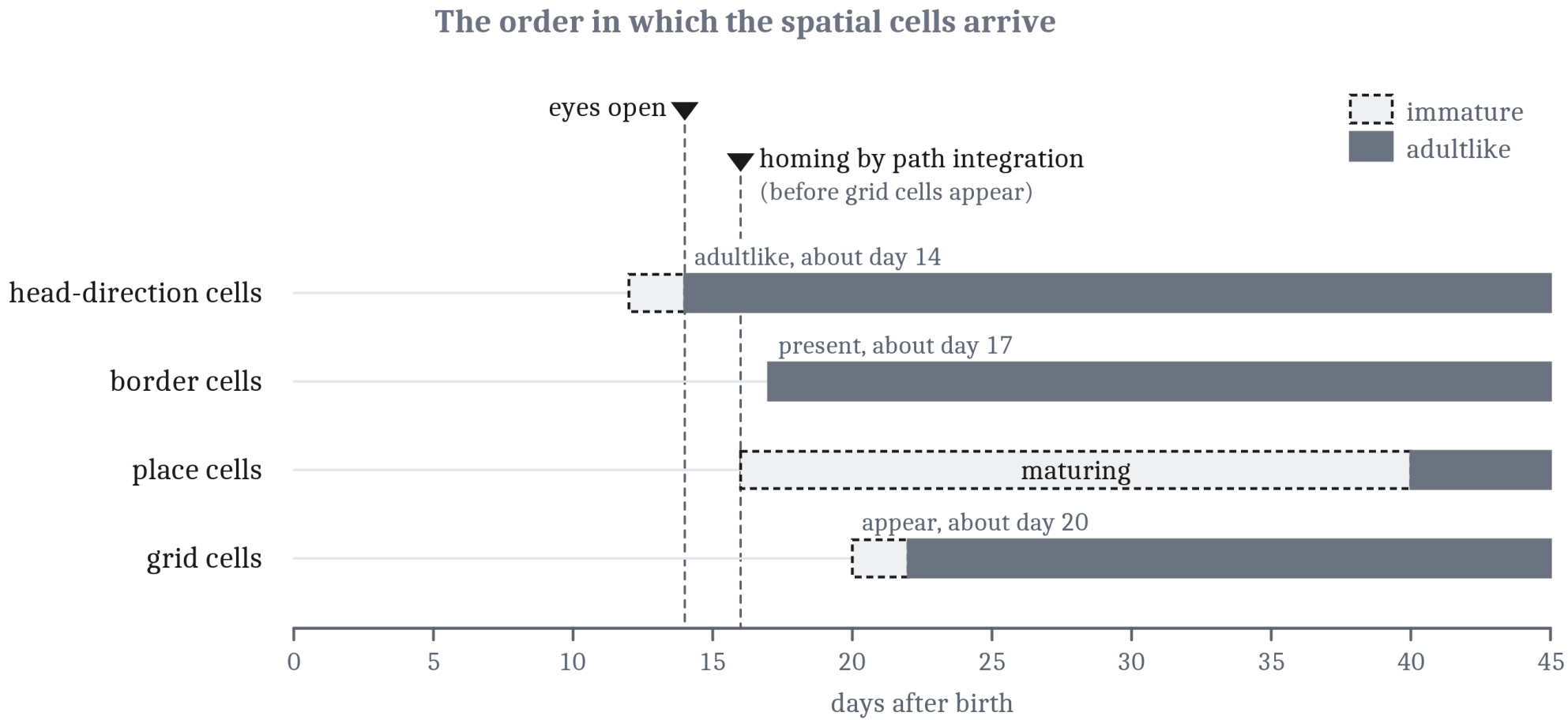
Head-direction cells are found in fish and birds, place cells in birds; grid cells so far only in mammals. As far as the evidence goes, mammals have an elaborated version of an inherited region.



Our own tree, after Striedter (2016) and Reiter and colleagues (2017); branch lengths are not to scale

Growing and Changing a Map

In rat pups, head-direction cells come first and grid cells last, after pups already home by path integration. In adults, trainee taxi drivers who qualified gained hippocampal gray matter, in a study that could not assign them at random



Redrawn from the data of Wills and colleagues (2010) and Wills and Cacucci (2026); approximate days

An Old Map in a New Animal

The early mammal did not need a new map. It needed an old one, put to harder use by a warm body that needed more food.

Computational

To reach a place it cannot sense, an animal needs to know which place is worth going to, where it is, where that place is, and which way to go.

Only knowing where places lie relative to one another allows a route never taken.

Building that knowledge means exploring: a trade-off with only rules of thumb.

Algorithmic

A running sum of position, corrected by stable landmarks.

Landmarks, routes, and the shape of the surroundings, each correcting the others.

A map, and a choice of which map one is on.

Exploring by rules of thumb, and curiosity.

Implementational

Goldfish, turtles, birds, and rats lose place-finding when a corresponding forebrain region is damaged, and keep cue-following.

One inherited region, reworked in each line: on the side of the forebrain in a goldfish; a dentate gyrus and a grid system in a rat.

One reviewer's summary: what the hippocampus does has changed much less than how it is built.

Taxi Drivers, and Phones

In your group, four minutes.

Many people now find their way by following directions on a phone.

If the taxi-driver result holds, what would you predict for the hippocampus of someone who always navigates that way? What study would show it, and what would it need to rule out?

Image Sources (1 of 2)

Slide	Image	File	License	Source
3	Head-direction cells, turned by a card	fig-hd-cells.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
4	The fruit fly's compass	fig-fly-compass.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
4	The larval zebrafish's heading ring	fig-two-rings.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading, adapted from Figure 6e of Petrucco and colleagues, Nature Neuroscience 26, 2023 · https://doi.org/10.1038/s41593-023-01308-5
5	Four kinds of spatial cell	fig-firing-maps.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
6	Grid cells in groups of different spacing	fig-grid-groups.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
7	The hippocampal loop in the rat	fig-hippocampal-circuit.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
8	A square box changed step by step into a circle	fig-morph-remapping.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
8	A grid cell and a place cell in two rooms	fig-remap-grid-phase.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading, adapted from Figure 6A of Whittington and colleagues, Cell 183, 2020 · https://doi.org/10.1016/j.cell.2020.10.024

Image Sources (2 of 2)

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9	The object-in-context test	fig-object-in-place.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
10	Goldfish before and after damage to the lateral pallium	fig-goldfish-place-cue.png	CC BY 4.0	Jon Willits — redrawn for the Module 9 reading from the values in Figure 4A of Broglio and colleagues, Journal of Behavioral and Brain Science 5, 2015, itself modified from Rodríguez and colleagues, Journal of Neuroscience 22, 2002 · https://doi.org/10.4236/jbbs.2015.53011
11	Evidence for a hippocampal map across vertebrates	fig-vertebrate-map-tree.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
12	When the spatial cells appear in rat pups	fig-pup-timeline.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading, from the data of Wills and colleagues (2010) and Wills and Cacucci (2026) · own work
13	Finding the way, at three levels	fig-old-map.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work